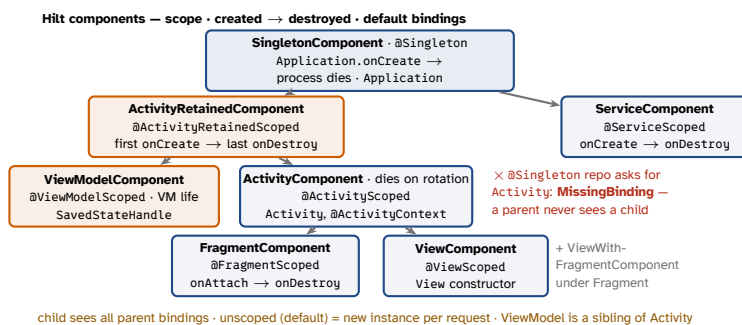


In one line: Dagger turns `@Inject`/`@Provides`/`@Binds` into a **dependency graph checked at compile time** and emits plain factory code — no reflection. Hilt fixes the **component tree** to Android lifetimes: a binding lives in one component, is visible to that component's **children**, and a **scope** annotation means “one instance per component instance”.



Build time — what the processors emit

```
class Repo @Inject constructor(api: Api) → Repo_Factory
@Provides fun api(ok: OkHttpClient): Api → NetModule_ApiFactory
@AndroidEntryPoint class MainActivity → superclass swapped to Hilt_MainActivity (Gradle plugin bytecode rewrite) → injects in onCreate
scoped: DoubleCheck.provider(Repo_Factory...) — lazily created once, cached in the component (locked); unscoped: factory.get() each time
graph validated per build: missing / duplicate binding, cycle, scope mismatch = compile error, not a crash. Processors: KSP (2.48+) or kapt
```

Koin: the same graph built by lambdas at **runtime** — errors surface on first use

Binding sources

- `@Inject` constructor — your own classes; prefer it.
- `@Provides` — code for types you don't own (Retrofit, Room builder). `@Binds` — abstract, **no body**, interface → impl, cheapest (no module instance).
- `@Module @InstallIn(X::class)` decides the component (and so who can see it and how long a scoped instance lives). A scope on a binding must match its component.
- Qualifiers** for two bindings of one type: `@Named("auth")` or a custom `@Qualifier` annotation class; built-ins `@ApplicationContext`, `@ActivityContext`.
- Multibindings**: `@IntoSet`/`@ElementsIntoSet` → `Set<T>`; `@IntoMap` + `@StringKey`/`@ClassKey`/custom `@MapKey` → `Map<K, V>` or `Map<K, Provider<V>>`; `@Multibinds` declares a possibly empty one. Plug-ins, interceptors, initialisers.
- `Provider<T>`: `get()` each call (new if unscoped). `Lazy<T>` (dagger.Lazy): built on first `get()`, then cached *per Lazy*. Both defer work and break a cycle.

Example

```
@Qualifier annotation class Auth
@Module @InstallIn(SingletonComponent::class)
abstract class DataModule {
    @Binds abstract fun repo(impl: NoteRepoImpl): NoteRepo
    @Binds @IntoSet abstract fun log(i: LogInterceptor): Interceptor
    companion object {
        @Provides @Singleton @Auth
        fun ok(set: Set<JvmSuppressWildcards Interceptor>) =
            OkHttpClient.Builder().apply { set.forEach { addInterceptor() }.build() }
    }
    @HiltViewModel(assistedFactory = NoteVM.Factory::class)
    class NoteVM @AssistedInject constructor(@Assisted val id: Long,
        repo: NoteRepo, heavy: Lazy<Parser>) : ViewModel() {
        @AssistedFactory interface Factory { fun create(id: Long): NoteVM }
    }
    // Compose: hiltViewModel<NoteVM, NoteVM.Factory> { it.create(id) }
```

Traps

- ! Scoping “for performance” — a scope is a lifetime + a lock, not a speed-up.
- ! `Set<Interceptor>` in Kotlin needs `@JvmSuppressWildcards`, or `MissingBinding`.
- ! Plain unit tests need no Hilt — call the `@Inject` constructor with fakes.

Remember: *Inject constructors, Bind interfaces, Provide the rest · install decides visibility · scope = one per component.*

Assisted injection & entry points

- `@AssistedInject` mixes runtime values (`@Assisted id`) with graph deps via a generated `@AssistedFactory`; VMs: `@HiltViewModel(assistedFactory = ...)` (Dagger **2.49**). Assisted values are **not saved** on process death — nav args go in `SavedStateHandle`.
- `@EntryPoint` = a door *into* the graph for what Hilt can't construct (ContentProvider, library callbacks): interface `E { fun repo(): Repo } + EntryPointAccessors.fromApplication(ctx, E::class.java).repo()`.

Testing

- `@HiltAndroidTest` + `@get:Rule` val hilt = HiltAndroidRule(this) (first rule) + `hilt.inject()`, `HiltTestApplication`; a **fresh** `SingletonComponent` per test.
- Per class: `@UninstallModules(NetModule::class)` + a nested test module. Whole source set: `@TestInstallIn(replaces = [NetModule::class], ...)`. One field: `@BindValue @JvmField val api: Api = FakeApi()`.

Koin vs Hilt

Koin: module { single { Repo(get()) }; viewModel { VM(get()) } }, resolved at **runtime**: fast builds, KMP-ready, a missing binding crashes on first use (`verify()` test; Koin Annotations add KSP checks). Hilt: compile-time, Android/JVM only.

Errors you will read

- [Dagger/MissingBinding] X cannot be provided without an `@Inject` constructor or an `@Provides`-annotated method — wrong component/qualifier, no `@Binds`.
- [Dagger/IncompatiblyScopedBindings] (`@Singleton` → `@ActivityScoped`); [Dagger/DependencyCycle] → `Provider/Lazy`.
- “Hilt Activity must be attached to an `@HiltAndroidApp` Application” — no `android:name`.